

KALABE KEBEDE

Senior Software Engineer | Forward Deployed Engineer

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github.com/Kalab21 | Portfolio: developer-portfolio-iota-ten.vercel.app

PROFESSIONAL SUMMARY

Senior Software Engineer and Forward Deployed Engineer with 6+ years building and modernizing enterprise systems across banking, financial services, insurance, analytics, and AI-enabled applications. Strong in Java/Spring Boot, distributed systems, Kafka, SQL/data engineering, AWS, APIs, Python/FastAPI, and production delivery, with recent hands-on work in LLM/RAG systems, pgvector semantic retrieval, LangChain/LangGraph, AI evaluation, and brownfield modernization.

TECHNICAL SKILLS

Software & Distributed Systems: Java 8-21, Spring Boot, Spring Security, Spring Data JPA, REST APIs, Microservices, Apache Kafka, Event-Driven Architecture, API Gateway, Concurrency, Idempotency

AI / GenAI & FDE: Python, FastAPI, LLMs, LLM Integration, RAG, RAG Architecture, Vector Databases (pgvector), Embeddings & Similarity Search, LangChain, LangGraph, LLM Pipelines, AI Evaluation, Human-in-the-Loop, Brownfield Analysis, Requirements Synthesis, ADRs

Cloud & Delivery: AWS, Amazon Bedrock, Docker, Kubernetes, Terraform, Infrastructure as Code (IaC), GitHub Actions, Jenkins, Maven, CI/CD

Data Engineering & Analytics: Advanced SQL, PostgreSQL, Oracle, MySQL, MongoDB, Redis, ETL/ELT, Data Pipelines, Data Quality, Query Optimization, Indexing, Data Modeling, Power BI, Tableau

Security, Reliability & Testing: OAuth2/OIDC, JWT, RBAC, Logging, Metrics, Tracing, Reconciliation, JUnit 5, Mockito, Pytest, Playwright, Integration Testing, TDD

PROFESSIONAL EXPERIENCE

Revature | Forward Deployed Engineer | May 2026 - Sep 2026

Remote

- Modernized inherited multi-service lending and assessment platforms by tracing workflows across APIs, services, databases, authorization boundaries, and operational controls, then delivering verified changes without breaking existing contracts.
- Built and hardened a grounded staff-only RAG/LLM assistant with Python/FastAPI, LangChain/LangGraph, bounded tool use, AI evaluation, and human review while keeping credit decisioning separate and authoritative in the core lending workflow.
- Translated business and regulatory requirements into specifications and ADRs; strengthened RBAC, decision finality, idempotency, auditability, reconciliation, logging, metrics, tracing, and CI/CD.
- Completed participant/trainer, scoring, reporting, analytics, and role-aware workflows across PostgreSQL/MongoDB-backed services and Nginx/API-gateway routing.

Lovelytics | Data & Analytics Engineer | Dec 2024 - Nov 2025

Remote

- Translated stakeholder reporting needs into SQL-based data models, KPI definitions, dashboards, and maintainable analytics workflows across PostgreSQL/MySQL and AWS data services.
- Built ETL/ELT, validation, and data-quality patterns and optimized analytical SQL for reliable downstream reporting and decision support.

Scale AI / Outlier AI | AI Data Specialist - Flexible / Task-Based Contract | Apr 2024 - Feb 2025

Remote

- Validated and refined AI-generated datasets and responses for factual accuracy, semantic quality, instruction adherence, and alignment with task requirements and business logic.
- Performed reviewer-level quality checks on peer submissions and provided targeted feedback to improve consistency and output quality.

PROFESSIONAL EXPERIENCE - CONTINUED

JPMorgan Chase | Java Software Engineer - Contract | May 2024 - Sep 2024

Columbus, OH | Hybrid

- Modernized enterprise banking services with Java and Spring Boot, improving API/database efficiency, maintainability, and deployment reliability in a compliance-sensitive environment.
- Built secure microservices and integrations using Spring Security, Spring Data JPA, Oracle, Redis, AWS, REST APIs, SFTP, and SMTP; improved reliability through query tuning, caching, automated tests, Docker/Kubernetes, CI/CD, and production support.

Investors Bank | Java Software Engineer | Apr 2021 - Dec 2023

Remote

- Modernized legacy core-banking workflows into Spring Boot microservices and Kafka-based event-driven services using PostgreSQL and Redis while preserving transaction and data integrity.
- Improved backend/database performance through SQL refactoring, indexing, caching, asynchronous processing, CI/CD, and distributed-system troubleshooting.

Erie Insurance | Java Software Engineer | Mar 2019 - Feb 2021

Erie, PA | Hybrid

- Developed Java backend services and REST APIs for claims, policy, customer, and transaction-processing workflows.
- Implemented concurrency-safe processing, Hibernate/JPA persistence, MySQL query optimization, automated testing, and production troubleshooting for business-critical services.

SELECTED ENGINEERING PROJECTS

Northbank Banking Platform - Personal Flagship | Distributed Systems

github.com/Kalab21/banking-platform

- Java 17/Spring Boot platform with 11 business services plus gateway/Eureka, Kafka, PostgreSQL, Redis, Next.js, Docker, transaction-safety controls, Terraform-based AWS topology, and 1,482 automated CI tests.

Meridian Lending - Team/FDE | Applied AI

github.com/2463-FDE/KK-meridian-lending

- Brownfield lending modernization with eight FastAPI backend services, decisioning, manual review, disclosures, servicing/payments, reconciliation, observability, and a staff-only LangChain/LangGraph RAG assistant.

Policy RAG Platform | Vector Search / RAG

github.com/Kalab21/policy-rag-platform

- Built a FastAPI RAG platform on PostgreSQL/pgvector: local 384-dimensional embeddings, HNSW cosine search, PostgreSQL full-text search with optional RRF hybrid retrieval, SQL metadata filtering, and a LangGraph evidence gate that returns cited answers or refuses.
- Added citation validation, a retrieval evaluation harness, and 168 automated tests including real PostgreSQL/pgvector integration tests; 0.939 MRR (semantic retrieval) on a small in-sample evaluation set.

Rev-Eval - Team/FDE | Full-Stack Platform Engineering

github.com/RevatureFDEPEP/rev-eval/tree/kalabek

- FastAPI/Next.js assessment platform with scoring/idempotency, trainer/participant workflows, reporting/analytics, JWT/RBAC, PostgreSQL, MongoDB, MinIO, Nginx, and CI/CD.

MarketHub - Personal | Java Full-Stack Engineering

github.com/Kalab21/markethub

- Spring Boot MVC marketplace with Spring Security, JPA/MySQL, Thymeleaf, Docker, GitHub Actions, and 39 automated tests.

EDUCATION & CERTIFICATION

B.S. Computer Science - Maharishi International University - 2023

B.S. Electrical & Computer Engineering - Haramaya University - 2015

Data Analytics Certificate - Per Scholas